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A APPENDIX

A.1 EVALUATION METRICS

Table 5: Evaluation metrics of ideas.

Metric	Definition
Novelty	Are the problems or approaches new? Is this a novel combination of familiar techniques? Is it clear how this work differs from previous contributions? Is related work adequately referenced?
Significance	Are the idea important? Are other people (practitioners or researchers) likely to use these ideas or build on them? Does the idea address a difficult problem in a better way than previous research? Does it provide a unique theoretical or pragmatic approach?
Clarity	Is the paper clearly written? Is it well-organized? Does it adequately inform the reader?
Feasibility	Can the idea be realized with existing technology or methods? Are there any technical difficulties or bottlenecks? Is the idea clear and logical? Is there any obvious error or unreasonable part in the idea, and can the experiment be designed normally according to this idea.
Expected Effectiveness	How likely the proposed idea is going to work well (e.g., better than existing baselines).

Table 6: Evaluation metrics of experiment design.

Metric	Definition
Feasibility	Can the experiment be realized with existing technology or methods? Are there any technical difficulties or bottlenecks? Is the experimental plan detailed and feasible? Are the experimental steps clear and logical? Is there any obvious error or unreasonable part in the experiment. Consider the rationality of its steps and the possibility that the idea can be successfully implemented.
Quality	Is there a clear rationale for each step of the experimental design? Are the baseline and evaluation metrics chosen appropriately? Has the design taken into account the potential advantages and limitations of the methods used? Can this experimental design effectively support the claims made in the idea.
Clarity	Is the experimental plan clearly written? Does it provide enough information for the expert reader to understand the experiment? Is it well organized? Does it adequately inform the reader?

A.2 SPECIFIC PROMPTS

Here are the prompts used in this paper.

- Prompts used in CoI construction
 - Prompt used to convert a topic into a search query for literature retrieval (Table 7)
 - Prompt used to evaluate whether a paper is relevant to the topic (Table 8)
 - Prompt used to extract idea, experiment, entities and references from paper (Table 9) and 10
 - Prompt used to summarize current trends of CoI (Table 11)
- Prompts used in idea generation

- Prompt used to predict future trend (Table 12 and 13)
- Prompt used to generate idea (Table 14 and 15)
- Prompt used to check the novelty of the idea (Table 16)
- Prompts used in experiment design
 - Prompt used to generate experiment design (Table 17)
 - Prompt used to review experiment design (Table 18)
 - Prompt used to get queries for search paper to refine experiment design (Table 19)
 - Prompt used to refine experiment (Table 20)
- Prompts used in benchmark construction
 - Prompt used to extract topic from real paper (Table 21)
 - Prompt used to extract the idea from real paper (Table 22)
 - Prompt used to extract the experiment design from real paper (Table 23)
- Prompts used in idea arena
 - Prompt used to compare two ideas (Table 24)
 - Prompt used to compare two experiment designs (Table 25)

Table 7: Prompt used to convert a topic into a search query for literature retrieval

You are a master of literature searching, tasked with finding relevant research literature based on a specific topic.

Currently, we would like to study the following topic: **[Topic]**
 Please provide the literature search queries you would use to search for papers related to the topic and idea.
 Each query should be a string and should be enclosed in double quotes. It is best to output one query representing the whole and other queries representing different aspects of the whole.

Output strictly in the following format:
 Queries: ...

A.3 ADDITIONAL EXPERIMENT RESULTS

We present the evaluation results of idea generation for both model-based evaluation (including GPT-4o, Gemini-1.5-Pro-Exp-0827, and Claude-3.5-Sonnet) and human-based evaluation in Table 26.